DEWDRIVE · PROBLEM STATEMENT

DewDrive problem statement

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.8	2026-10-25	Amish Chadha	Kit 1.8 pipeline: users-and-needs table and stakeholders table (STANDARDS section 22); cost figure from DWD-CAL-001 v0.8 after the STEM check fixes
0.7	2026-10-02	Amish Chadha	First partner type and region decided on 2026-10-02 (DWD-DEC-001, item 5)
0.6	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.5	2026-09-26	Amish Chadha	Budget top-up approved by Amish
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002); yield and cost figures from DWD-CAL-001 v0.2
0.3	2026-09-25	Amish Chadha	TRL 3 update (budget and problem wording decided by Amish in DWD-DDR-001; yield figures from DWD-CAL-001; partner question still open)
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (problem, users, context, constraints, out of scope,

			prior work with sources, open questions)
0.1	2026-09-24	Amish Chadha	Initial scaffold

DewDrive problem statement

Arid households without surface water or groundwater within reach depend on carried or trucked water; the air above them is an untapped supplementary source. A small, open, solar-powered harvester could turn that moisture into a daily supplement of drinking water, but the physics limits it to well under a litre per square metre per day in dry air (DWD-CAL-001 v0.8 estimates about 0.44 L per day for the design decided on 2026-09-25, with night air drawn through the sorbent trays and, after the STEM check of 2026-10-25, black absorber plates on the beds and an open tray floor), so the realistic aim is a supplementary drinking-water source that a household or school can build and repair, not a replacement for a well or a water truck. Design with, not for: requirements must come from co-design sessions and field trials with the intended users through a local partner.

The problem

Where there is no river, spring or aquifer within reach, water is carried from far away, bought from vendors or trucked in, and it is often of uncertain quality. The World Health Organization puts survival needs for drinking and food at 2.5 to 3 L per person per day, and basic needs including hygiene and cooking at 7.5 to 15 L (WHO technical note 9).

Even desert air holds water. At 20 °C and 40 % relative humidity (RH), each cubic metre carries about 7 g of vapour. Field trials in Scottsdale, Arizona, recorded night RH of about 40 % falling to 8 % by day (Fathieh et al., *Science Advances*, 2018; ScienceDaily summary). Cooling air below its dew point, as a fog net or a refrigeration dehumidifier does, fails at this humidity or costs too much energy. A desiccant can instead soak up vapour from cool, relatively humid night air, and the sun can drive it out again by day into a condenser.

Three gaps keep this from reaching the people who need it:

- **Cost and closed designs.** Commercial solar hydropanels produce about 3 to 5 L per day and cost about $2,000 each (Forbes, 2022). They are sealed products that cannot be built or repaired locally.
- **Exotic sorbents.** Much of the research uses metal-organic frameworks (MOFs) or engineered zeolites that are not sold in small quantities at low cost.
- **Unclear real-world yield.** Published yields vary tenfold with humidity, sorbent and device design, so a household cannot tell whether a harvester would be worth building.

Users and context

The user requirements in DWD-URS-001 are drawn from Tables 1 and 2. Both are proposals until co-design sessions with the partner confirm them.

Table 1. Users and needs (to be confirmed through co-design).

User type	Goals and needs	Typical use scenario	Pain points and challenges
Rural household in an arid or semi-arid area	A daily supplement of safe drinking water, especially in the dry season, for little daily effort and no fuel	Opens the flaps at dusk and shuts them at dawn; weighs and empties the jerrycan; treats the water before drinking	Water carried or bought; nights of 25 to 40 % RH; cannot afford a commercial hydropanel; no grid; children near the device
School or clinic	Drinking water for children or patients when the delivered supply runs short	Staff run the twice-daily routine; the unit stands in a yard or on a flat roof	Staff time; device must be safe around children; water must be treated and tested
Local technician or maker	A design they can build from local materials and repair with common tools	Builds the box, stand and trays from the build plan; replaces the fan, battery or seals	Sheet-metal, woodwork and basic electronics skills; parts from general suppliers only; no special tooling
Researcher or NGO	Open, logged yield data across climates, and a device to compare sorbents	Installs several units at partner sites; downloads the logger each month	Yields in the literature vary tenfold; closed products give no data

Table 2. Stakeholders.

Stakeholder group	Role or interest	Engagement needed
First partner: university dryland field station (candidate type decided 2026-10-02, not yet agreed)	Hosts the first prototype, supplies the weather record and runs the TRL 4 tests	Agreement, site data, co-design of the user requirements
Water and sanitation NGO (later community trial)	Brings the household and school users, owns water-quality testing in the community	Co-design sessions, water testing and treatment plan
Local water and health authority	Sets drinking-water rules where the water is used	To assess per country (DWD-REG-001)
Local workshops and suppliers	Build, repair and supply the parts	Check of the build plan and part availability
Amish Chadha (Design Molecule)	Design owner; decides scope, budget and safety trade-offs	Decisions recorded in DWD-DEC-001

Operating environment (proposed design envelope):

- Night air 10 to 30 °C at 25 to 70 % RH; day air 25 to 45 °C at 5 to 30 % RH.
- Clear-sky solar irradiation about 5 to 7 kWh/m² per day on a tilted surface (estimate for desert sites, to be confirmed with site data).
- Dust, wind to 20 m/s (72 km/h), ultraviolet exposure, animals and occasional rain.
- No grid power. The site may be visited only twice a day.

Constraints

- Garage-buildable research prototype, a value-engineering target of $520 USD in parts (`project.yaml` ; a hypothetical control target, not a limit; moved from $400 to $500 by Amish on 2026-09-25, DWD-DDR-001, D1,

and to $520 on 2026-09-26, DWD-DDR-002, D15). The TRL 3 estimate was $515, within the target; after the design for construction it was $639, and after the STEM check fixes of 2026-10-25 it is $695, $175 over the target (DWD-CAL-001 v0.8).

- Regeneration heat from the sun only. Any electricity (fan, logger) comes from the device's own small solar panel.
- Common, low-cost sorbents: silica gel and calcium chloride, both sold as desiccants and food-grade chemicals.
- Everything in contact with the water must be food-grade.
- Two people can carry and set it up without tools beyond a spanner and a screwdriver.
- Open hardware under CERN-OHL-S-2.0; software under MIT.

Out of scope

- Providing all of a household's water. One unit is sized as a supplement (see DWD-PRS-001).
- Refrigeration or compressor-based dehumidifiers.
- MOF or other specialty sorbents in the first prototype (possible later upgrade).
- Treating the water to a certified potable standard. The design keeps the water clean in the device, but the documents require testing and treatment before drinking until tests show otherwise.

Prior work

- **MOF harvesters.** Kim and colleagues showed solar-driven harvesting with MOF-801 at 20 % RH (*Science*, 2017). The Arizona field trial produced about 0.2 L per kg of MOF-801 per day, and the aluminium-based MOF-303 was expected to double that at far lower material cost (Fathieh et al., 2018).
- **Dual-stage zeolite device.** LaPotin and colleagues at MIT reached about 0.77 L/m² per day outdoors with a dual-stage AQSOA Z01 device, at about 9 % thermal efficiency and absorber temperatures up to 94 °C, with overnight air at about 68 % RH (*Joule*, 2021).
- **Salt-in-silica composites.** Calcium chloride confined in mesoporous silica gel (SWS-1L) holds up to 0.6 to 0.7 g of water per gram (Aristov et al., *Chemical Engineering Science*, 2006). A passive harvester with a lithium chloride and silica gel composite produced 355 g per day from 0.48 m² (about 0.74 L/m² per day), with 0.52 g/g uptake at 30 °C and 60 % RH (Shao et al., *Energy*, 2024).
- **Salt in a polymer matrix.** An alginate-derived $CaCl_2$ composite took up its own weight in water at 10 mbar vapour pressure and released it at 100 to 150 °C (Kallenberger and Fröba, *Communications Chemistry*, 2018).
- **Commercial hydropanels.** SOURCE hydropanels produce about 3 to 5 L per day per panel at about $2,000 each (Forbes, 2022).

The gap DewDrive targets is an open, documented and locally buildable device using the low-cost composite route, with logging built in so that yield can be compared between sites.

Open questions

- [ ] What are the real night humidity and solar figures at the partner's site? The partner and region were decided by Amish on 2026-10-02 (DWD-DEC-001, item 5): the first candidate type to approach, not yet agreed, is a university dryland field station in a hot semi-arid region with night humidity of about 25 to 40 % and an existing weather record, with a water and sanitation NGO for the later community trial.
- [ ] Is about 0.5 L per day per unit worth a household's time and money, compared with carrying or buying water? DWD-CAL-001 v0.8 puts the parts cost at about $0.87 per litre over five years.

- [ ] How will households test and treat the water, and who pays for the tests?
- [ ] Would a school or clinic, with staff on site twice a day, be a better first user than a household?

> **Safety:** Harvested water must be tested and treated before drinking. Calcium chloride is an eye and skin irritant, and the box interior runs hot enough to burn. See the safety section of DWD-PRC-001.